ModelIt!

Student Activity Pack

Is Your Tap Water Safe?

Modeling Water Contamination Pathways and Public Health Systems

Grade 12L1 | NGSS HS-PS1-2, HS-ESS3-4 | Systems and System Models

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Where does the water in your tap come from, and what happens to it before it reaches your glass?**

2. **Have you ever heard of a city having contaminated drinking water? What do you know about what happened?**

3. **What do you think is in your tap water right now besides H2O?**

4. **Who is responsible for making sure your drinking water is safe?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Source Water Quality | Treatment Effectiveness | Pipe Infrastructure Age | Corrosion Rate | Tap Water Safety

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Healthy System

Source: Clean | Treatment: Full | Pipes: Modern copper/PVC

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Flint Scenario

Source: Corrosive river water | Treatment: No corrosion control | Pipes: 1920s lead

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Aging Infrastructure

Source: Clean | Treatment: Full | Pipes: 80+ year old lead/iron

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Testing Frequency — How often water quality is measured at various points in the distribution system — less frequent testing means contamination can go undetected for weeks or months

Community Socioeconomic Status — The economic resources of the community served — lower-income communities often have older infrastructure, less political power to demand repairs, and fewer resources for bottled water alternatives

Regulatory Enforcement — The strength and consistency of government oversight — weak enforcement allows utilities to delay maintenance, skip tests, or manipulate results

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

The water coming out of your faucet traveled through miles of pipes before reaching your glass — how do we know what's in it, and what happens when the system fails? Use evidence from your simulation to support your answers.

1. **How does your model explain why the Flint water crisis was a systems failure rather than a single-point failure?**

2. **Why might lower-income communities be more vulnerable to water contamination than wealthier ones?**

3. **What does your model suggest about the relationship between infrastructure investment and public health?**

4. **If you could add one component to your model to make it more accurate, what would it be and why?**

5. **How does the concept of 'parts per billion' change the way you think about contamination?**

8 STEM Challenge: Design a Community Water Monitoring System

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YOUR ENGINEERING MISSION

Design a low-cost, community-based water quality monitoring system that can detect contamination before it reaches dangerous levels and alert residents in real-time.

THE SCENARIO

After the Flint crisis, a neighboring city wants to prevent a similar disaster. They have aging infrastructure, limited budget for pipe replacement, and want residents to be empowered with water quality data. Your team must design a monitoring system that provides early warning of contamination events.

Engineering Design Thinking

1. WHAT SPECIFIC SENSORS AND TESTING METHODS DOES YOUR SYSTEM USE AND WHAT IS THE SCIENTIFIC BASIS FOR EACH?
2. HOW DOES YOUR MONITORING NETWORK COVERAGE RELATE TO THE DISTRIBUTION SYSTEM ARCHITECTURE?
3. WHAT ALERT THRESHOLDS TRIGGER ACTION AND HOW WERE THEY DETERMINED FROM MODEL DATA?
4. HOW WOULD YOUR SYSTEM HAVE DETECTED THE FLINT CRISIS EARLIER THAN THE ACTUAL TIMELINE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Environmental engineers design water treatment systems and monitor water quality, earning $65,000-$110,000/year. Water quality scientists test drinking water for contaminants at utilities and government agencies, earning $50,000-$85,000/year. Environmental justice advocates work to ensure equitable access to clean water, earning $45,000-$80,000/year.